

Transplantation

Addis Ababa University
Clinical Pharmacy

Definition

- Transplantation refers to
 - the surgical procedure of transferring cells, tissues, or organs from a donor to a recipient to restore function lost due to disease, injury, or congenital defects.
 - In clinical practice, the term typically refers to solid organ transplantation (kidney, liver, heart, lung, pancreas) and hematopoietic stem cell transplantation (HSCT).

Etiology / Indications

- Transplantation is indicated when irreversible organ failure occurs, including:
 - End-Stage Renal Disease (ESRD) — kidney transplant
 - End-Stage Liver Disease (ESLD) — cirrhosis, hepatitis
 - Heart Failure — advanced cardiomyopathy
 - Lung Diseases — COPD, pulmonary fibrosis
 - Pancreas — type 1 diabetes with complications
- Organ failure may be
 - acquired (e.g., diabetes, hypertension) or
 - congenital (e.g., cystic fibrosis for lung).
- Hematopoietic Stem Cell Transplantation (HSCT) is used for malignant (leukemia) and non-malignant (aplastic anemia) hematologic conditions.

Epidemiology

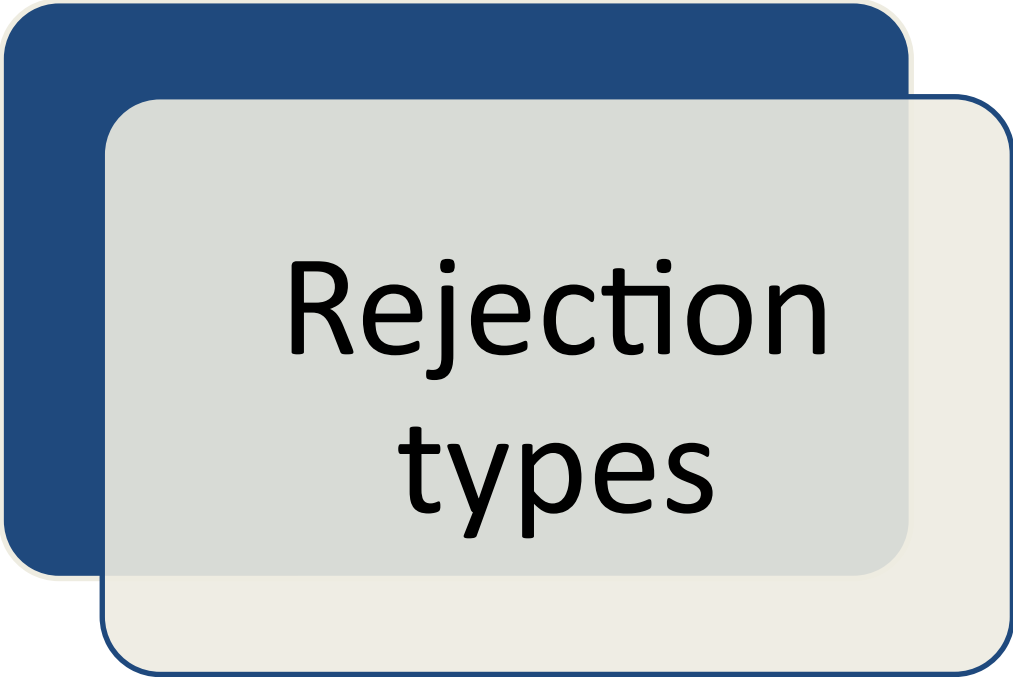
- Globally, **kidney transplantation** is the most common solid organ transplant.
- Graft survival and patient survival have improved dramatically over decades, largely due to advances in immunosuppression and supportive care.
- Contemporary outcomes are:
 - 1-year graft survival > 90% for many organs
 - Long-term survival limited primarily by chronic rejection, immunosuppression complications, cardiovascular disease, and infections.
- Epidemiology varies by region, organ, and access to transplant programs, with disparities in donor availability and healthcare infrastructure.

Clinical Picture

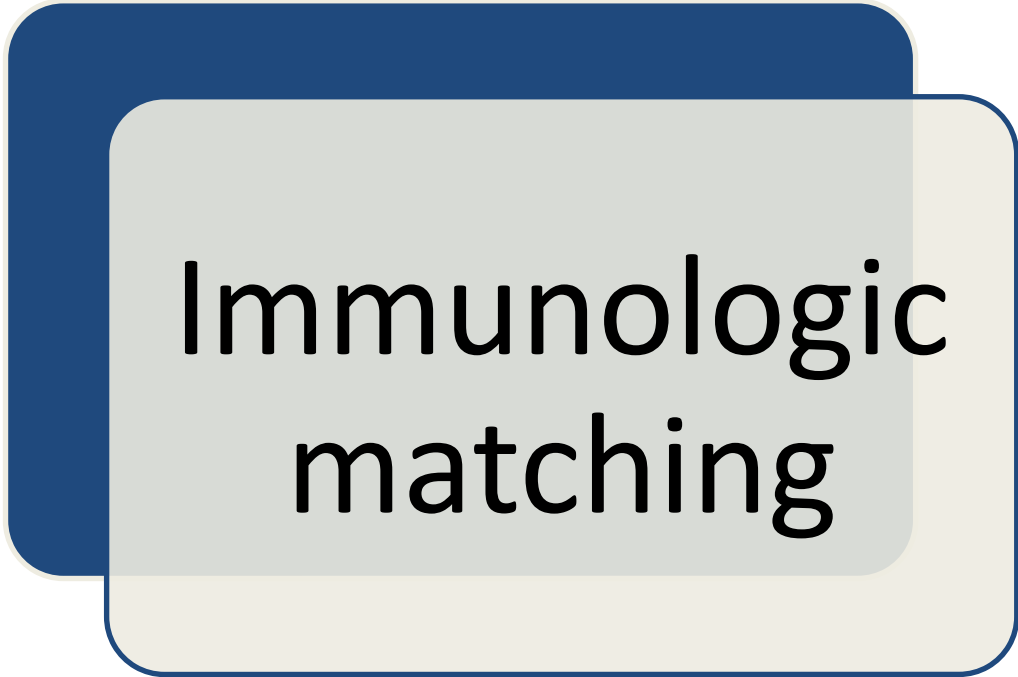
- **Normal Post-Transplant Course**
 - Immediate postoperative recovery
 - Gradual graft function (e.g., kidney urine output)
 - Long-term immunosuppressive therapy
- **Complications**
 - **Rejection**
 - *Hyperacute* — within minutes (preformed antibodies)
 - *Acute* — days to months (cellular/humoral immunity)
 - *Chronic* — months to years (vascular and interstitial fibrosis)
 - **Infections** — opportunistic infections due to immunosuppression
 - **Drug toxicity** — nephrotoxicity, metabolic complications
 - **Malignancy** — due to chronic immunosuppression



Transplantation Immunology



Rejection
types



Immunologic
matching

Rejection & Monitoring

- Fever, graft tenderness, decreased organ function (e.g., rising creatinine in kidney).
- Laboratory
 - Allograft function markers (creatinine, liver enzymes, etc.)
 - Therapeutic drug levels (Tacrolimus, Cyclosporine)
- Histopathology
 - **Biopsy** remains gold standard for acute and chronic rejection — grading by specific criteria (e.g., Banff for kidney).

Goals of Transplantation

- The primary goals of immunosuppressive pharmacotherapy are to:
 - Prevent rejection — acute and chronic
 - Minimize drug-related toxicity
 - Maintain graft and patient survival
 - Reduce infection and malignancy risk
 - Achieve long-term allograft function with acceptable quality of life

Transplantation Types & Classification

- By Donor Type
 - Living donor — genetically related/unrelated
 - Deceased donor
 - Expanded criteria donors
- By Immunologic Risk
 - High risk — sensitized patients with preformed antibodies
 - Low risk — first transplant, compatible match
- By Immunosuppressive Strategy
 - Induction therapy — intense, short-term after transplant
 - Maintenance therapy — long-term
 - Antirejection therapy — for acute episodes.

*Immunosuppressive regimens are often described as triple therapy (calcineurin inhibitor + antiproliferative + corticosteroid).

Drugs Used During Transplantation

A. Induction Agents (Short-Term)

Agent	Class/Target	Role
Basiliximab	IL-2 receptor antagonist (anti-CD25)	Prevent acute rejection at transplant time
Antithymocyte Globulin (ATG)	Polyclonal antibody depleting T cells	Potent induction in high-risk patients
Muromonab-CD3	Anti-CD3 monoclonal	Historical use; limited now due to side effects
Alemtuzumab	Anti-CD52 monoclonal	Used in some protocols

Drugs Used During Transplantation...

B. Maintenance Immunosuppression (Long-Term)

Class	Examples	Mechanism / Notes
Calcineurin inhibitors	Tacrolimus, Cyclosporine	Inhibit IL-2 transcription and T-cell activation
Antiproliferative agents	Mycophenolate mofetil, Azathioprine	Inhibit lymphocyte proliferation
mTOR inhibitors	Sirolimus, Everolimus	Block T-cell proliferation by mTOR pathway
Corticosteroids	Prednisone	Broad immunosuppression & anti-inflammatory
Costimulation blockers	Belatacept	Blocks CD80/86 costimulation

Drugs Used During Transplantation...

C. Antirejection Therapy

Used when rejection occurs:

- High-dose corticosteroids
- Antithymocyte globulin
- Monoclonal antibodies (e.g., alemtuzumab, rescue therapies)
- IV immunoglobulin (IVIG) and plasmapheresis for antibody-mediated rejection

Outcomes of Transplantation

Short-Term

- Improved graft survival due to effective induction and improved surgical techniques.
- Significant reduction in acute cellular rejection.

Long-Term

- Chronic rejection remains a cause of late graft loss.
- Immunosuppressive toxicity (nephrotoxicity, metabolic syndrome)
- Infection and malignancy due to immunosuppression.
- Ongoing research focuses on personalized immunosuppression and tolerance protocols.

Monitoring

- Clinical & Laboratory
 - Therapeutic drug monitoring (TDM) for CNIs (e.g., tacrolimus trough levels)
 - Organ function tests (creatinine, liver enzymes)
 - Infection surveillance
- Immunologic
 - Donor-specific antibody (DSA) monitoring
 - Biopsy for suspected rejection
- Side Effect Monitoring
 - Renal function
 - Blood pressure, glucose, lipids
 - Signs of infection or malignancy

References

- Varun Kalluri H, Hardinger KL. *Current state of renal transplant immunosuppression: Present and future*. World J Transplant. 2012.
- Hussaini T, et al. *Immunosuppressive pharmacotherapy in liver transplantation*. AME Med J.
- International J Basic & Clin Pharmacol. *Immunosuppressive drugs in renal transplantation*.

